Continuous Probability Distributions

1. Explore the shape of the normal distribution with an arbitrary mean and standard deviation in the following link:

   https://rexamples-golnaz.shinyapps.io/App-2/.

Possible considerations are:

- Change the value of mean and standard deviation to some value of your choice.
- Change the sample size to the value of your choice.
- Change the number of bins.

2. Now explore the probability of X using the normal distribution in the following link:

   https://rexamples-golnaz.shinyapps.io/App-3/

   Possible considerations are:

   - In the sidebar, change up the values and explore different probabilities.
   - Perhaps consider some examples from the lecture.

3. In R, obtain 10 random samples from the standard normal distribution (Hint: use the function rnorm(n = sample size, mean = 0, sd =1)).

   (a) Create a density histogram of this data and overlay the standard normal distribution curve. Increase your sample size (e.g. 100, 1000, 10,000). Comment on the shape of the distribution as you increase the sample size. Repeat this task for other distributions.

   e.g.

```r
# Generate 1000 random values from the standard normal
distribution
random_values <- rnorm(1000)

# Create a histogram of the random values
hist(random_values, main = "Histogram with Standard Normal
Distribution Curve", xlab = "X", probability = TRUE)

# Overlay a standard normal distribution curve
curve(dnorm(x), add = TRUE, col = "darkred", lwd = 2)
```

(b) Compute quantiles from the standard normal distribution with the following probabilities

i. 0.5
ii. 0.05
iii. 0.95
iv. 0.90
v. 0.10

Comment on your results. What do they correspond to?

(c) Compute probabilities from the standard normal distribution with the following X values

i. 0
ii. -1.64
iii. 1.64
iv. 1.28
v. -1.28

Comment on your results. What do they correspond to?

4. In R, solve the following

   (a) If $X \quad N(4,16)$ find $P(X>5)$;

   (b) If $X \quad N(0,25)$ find $P(X \leftarrow 1)$;

   (c) If $X \quad N(10,9)$ find $P(7.5<X<12.5)$;

   (d) If $X \quad N(3.9,0.81)$ find $P(X>3.1)$.

5. In R attempt the following question: If the average height is six feet tall and the standard deviation is half a foot.

   (a) what proportion of people are under five feet tall?

   (b) How likely is it that a person be between 5 foot 8 inches and 6 foot 3 inches tall?

   (c) According to this model, above what height are 20% of people?

6. The actual amount of coffee (in grams) in 230 g jar filled by a machine is a random variable, X, whose probability density function is given by

$$f(x)=\begin{cases}0 & x\leq 227.5\\ \frac{1}{5} & 227.5<x<232.5\\ 0 & x\geq 232.5\end{cases}$$

   (a) State what kind of distribution this is, and what its parameter values are.

   (b) Find the cumulative distribution function for X.

   (c) What is the probability that a 230g jar will contain at most 228.65g?

   (d) What is the probability that a 230g jar will contain between 229.34 and 231.66g?

7. Petrol is delivered to a garage every Monday morning. The weekly demand for petrol at this garage, in thousands of gallons, is a random variable, X, with the probability density function of the form

$$f(x)=\begin{cases}\frac{6}{5}x(3-2x) & \frac{1}{2}<x<\frac{3}{2}\\ 0 & \text{elsewhere}\end{cases}$$

(a) Calculate the mean weekly demand.

(b) Calculate the most likely demand for petrol.

8. The time, T, in seconds, between successive phone calls to arrive at a switchboard is given by

$$f(T=t)=2.32e^{-2.32t} \quad t\geq 0$$

(a) State what kind of distribution this is.

(b) Calculate the median time between successive phone calls to arrive at the switchboard.

Additional Exercises

1. The probability density function of the random variable *X* is given by

$$f(x)=\begin{cases}6x(1-x) & 0<x<1\\ 0 & \text{elsewhere}\end{cases}$$

(a) Find the cumulative distribution function for X.

(b) Find $P\left(X<\frac{1}{4}\right)$

(c) Find $P\left(X>\frac{1}{2}\right)$

2. The daily amount of coffee, in litres, dispensed by a machine located in an airport lobby is a random variable, *X*, having the following probability density function:

$$f(x)=\begin{cases}\frac{1}{3} & 7<x<10\\ 0 & \text{elsewhere}\end{cases}$$

(a) State what kind of distribution this is, and what its parameter values are.

(b) Find the probability that on a given day the amount of coffee dispensed by this machine will be

i. at most 8.8 litres;

ii. more than 7.4 litres but less than 9.5 litres;

iii. at least 8.5 litres.

3. At a certain location on motorway M25, the time (in hours) between two cars exceeding the speed limit by more than 10 miles per hour is a random variable having the following distribution:

$$f(x)=8.4e^{-8.4x} \quad x\geq 0$$

(a) State what kind of distribution this is

(b) What is the probability of a waiting time of less than 5 minutes between cars exceeding the speed limit by more than 10 miles per hour?